Tutorial 9 (Friday)

March 15, 2025   2:03 PM

Problem #1: Find the length of the following parametric curve.

$$x = 3 + \frac{15}{2}t^5, \quad y = 9 + 5t^{15/2}, \quad 0 \le t \le 2.$$

use: $\boxed{L = \int_\alpha^\beta \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt}$

$\alpha = 0, \ \beta = 2 \qquad \frac{dx}{dt} = \frac{15}{2}(5)t^4 \qquad \frac{dy}{dt} = 5\left(\frac{15}{2}\right)t^{\left(\frac{15}{2}-\frac{2}{2}\right)}$

$\frac{dx}{dt} = \frac{75}{2}t^4 \Rightarrow \left(\frac{dx}{dt}\right)^2 = \frac{75^2}{4}t^8$

$\frac{dy}{dt} = \frac{75}{2}t^{13/2} \Rightarrow \left(\frac{dy}{dt}\right)^2 = \frac{75^2}{4}t^{13}$

$L = \int_0^2 \sqrt{\frac{75^2}{4}t^8 + \frac{75^2}{4}t^{13}}\, dt = \int_0^2 \sqrt{\frac{75^2}{4}t^8(1+t^5)}\, dt$

$= \int_0^2 \frac{75}{2}t^4\sqrt{1+t^5}\, dt \qquad$ let $u = 1+t^5 \quad du = 5t^4 dt$

$dt = \frac{du}{5t^4}$

$= \int_1^{33} \frac{75}{2}\cancel{t^4}\, u^{1/2} \frac{du}{5\cancel{t^4}} \qquad$ and $t \to 0 \ u \to 1$, $t \to 2 \ u \to 33$

$= \frac{15}{2}\int_1^{33} u^{1/2} du = \frac{15}{2}\left(\frac{u^{3/2}}{3/2} + c\right)\Big|_1^{33}$

$= \frac{15}{2}\left(\frac{(33)^{3/2}}{3/2} - \frac{1^{3/2}}{3/2}\right) = \frac{15}{\cancel{2}}\frac{\cancel{2}}{3}\left(33^{3/2} - 1\right) = 5\left(33^{3/2} - 1\right)$

Problem #2: The Cartesian coordinates of a point are (1,−1). Find polar coordinates (r, θ) of the point, where

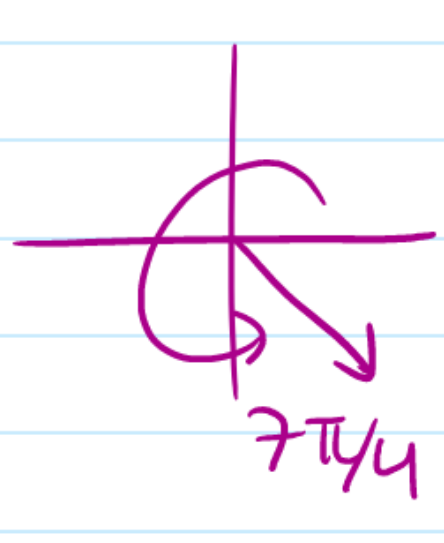
(a) $r > 0$ and $0 \le \theta < 2\pi$.
(b) $r < 0$ and $-2\pi \le \theta < 0$.

Part a)

$\boxed{x = r\cos\theta \quad y = r\sin\theta} \qquad \boxed{r = \sqrt{x^2+y^2}}$

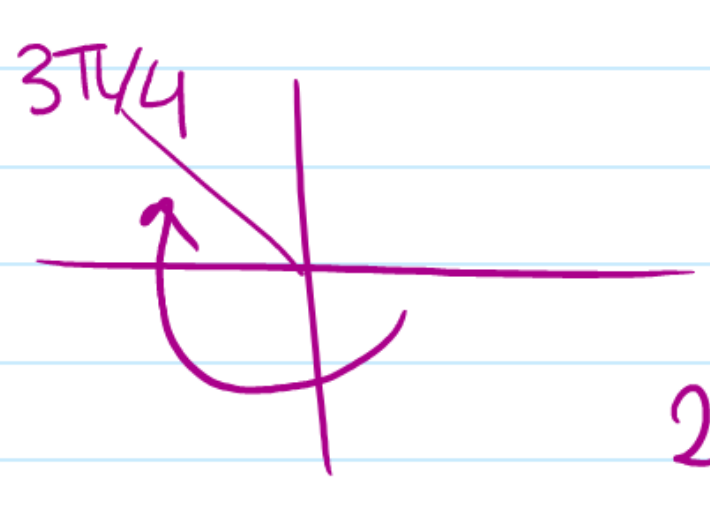
$1 = \sqrt{2}\cos(\theta) \quad -1 = \sqrt{2}\sin(\theta) \qquad = \sqrt{1+1}$

$1/\sqrt{2} = \cos(\theta) \quad -1/\sqrt{2} = \sin(\theta) \qquad = \sqrt{2}$



Hence, $(r, \theta) = (\sqrt{2}, 7\pi/4)$

Part b) $r = -\sqrt{2} \Rightarrow -1/\sqrt{2} = \cos(\theta) \quad 1/\sqrt{2} = \sin(\theta)$



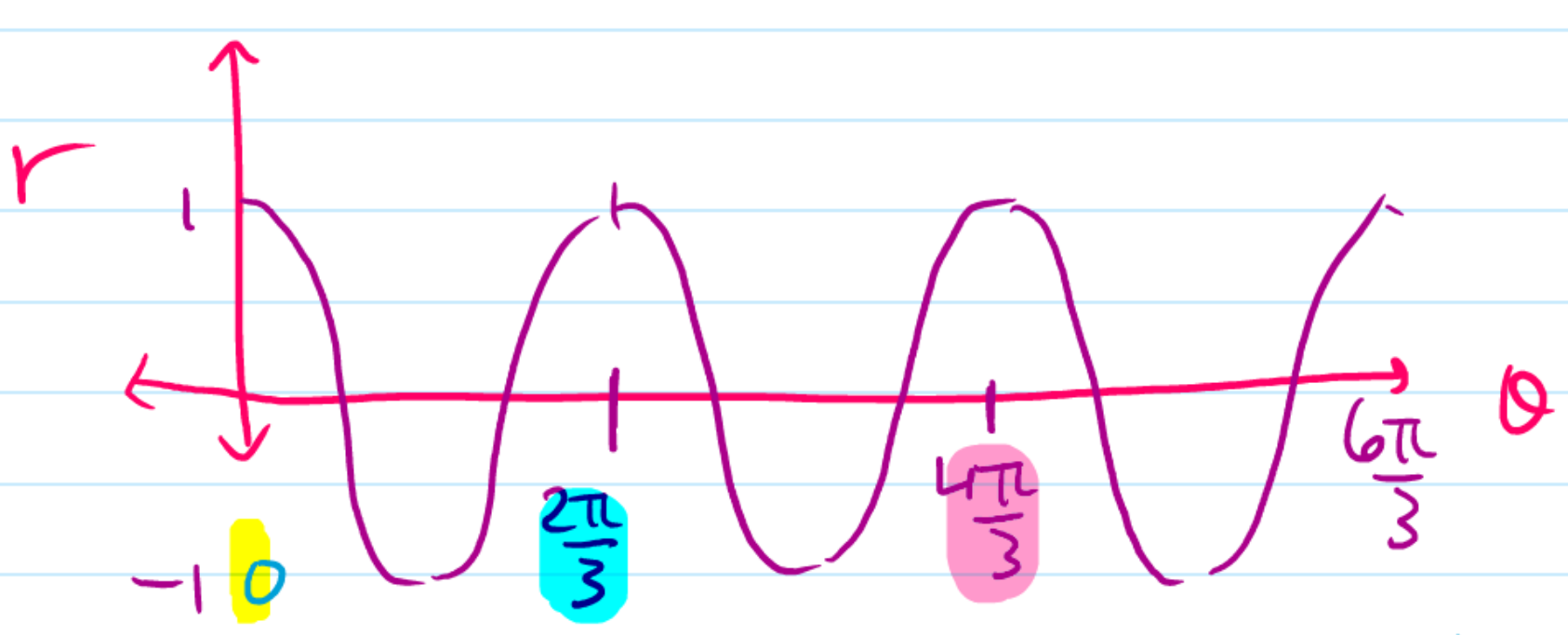
Hence, $(r, \theta) = (-\sqrt{2}, -5\pi/4)$

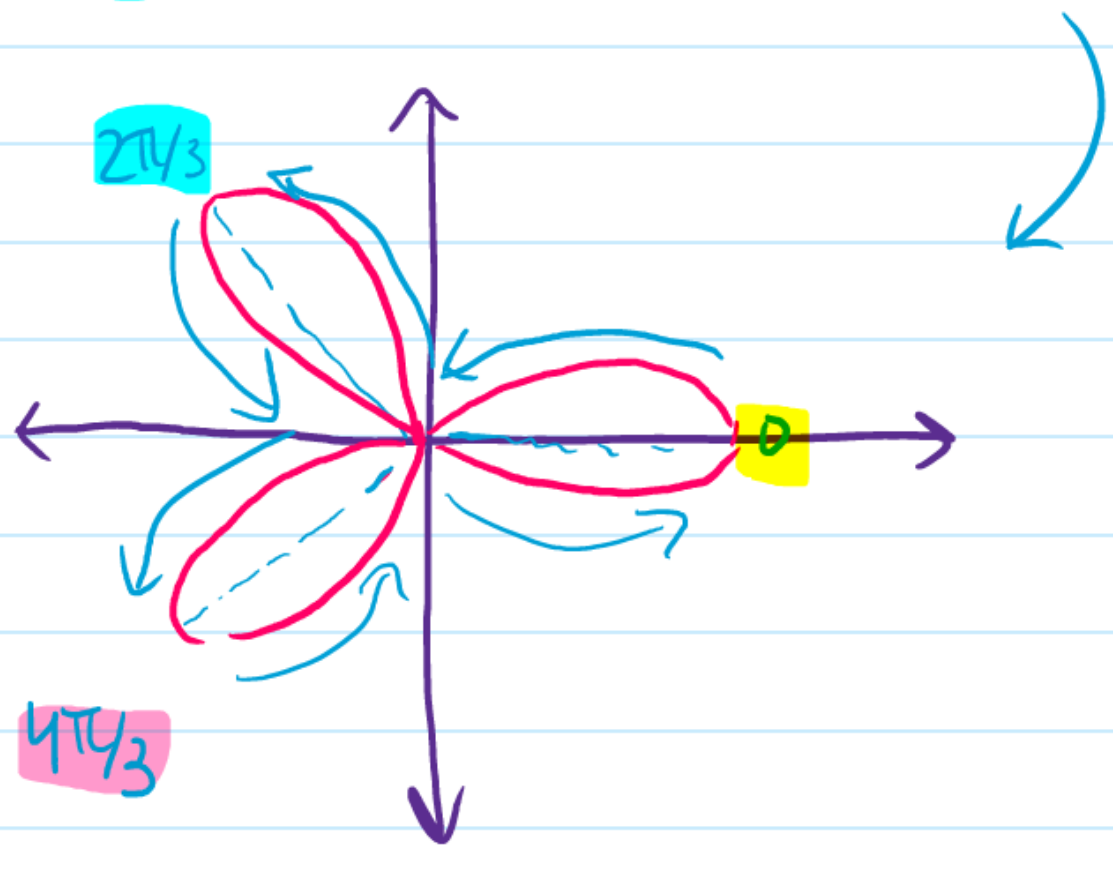
$2\pi - \frac{3\pi}{4} = \frac{8\pi - 3\pi}{4} = \frac{-5\pi}{4}$

Problem #3: Sketch the polar curve $r = \cos(3\theta)$. (you don't need the power of elimination).

1. Try to graph w/ Cartesian coordinates first.

$3\theta = 2\pi$ (first cycle)
$\theta = \frac{2\pi}{3}$





use as a guide.

Problem #4: Consider the polar graph $r = 2 + \cos(7\theta)$.

(a) Find the slope of the tangent line when $\theta = \frac{\pi}{2}$.
(b) The equation of the tangent line when $\theta = \frac{\pi}{2}$ is of the form $y = mx + b$. What is the value of b?

a) $r = 2 + \cos\left(7\frac{\pi}{2}\right) = 2 + \underbrace{\cos\left(\frac{3\pi}{2}\right)}_{0} = 2$

$\sin(\pi/2) = 1 \qquad \cos(\pi/2) = 0 \qquad r = 2 + \cos(7\theta)$

$\frac{dr}{d\theta} = -7\sin(7\theta)$

$\boxed{\frac{dy}{dx} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}} \qquad = -7\sin\left(7\frac{\pi}{2}\right)$

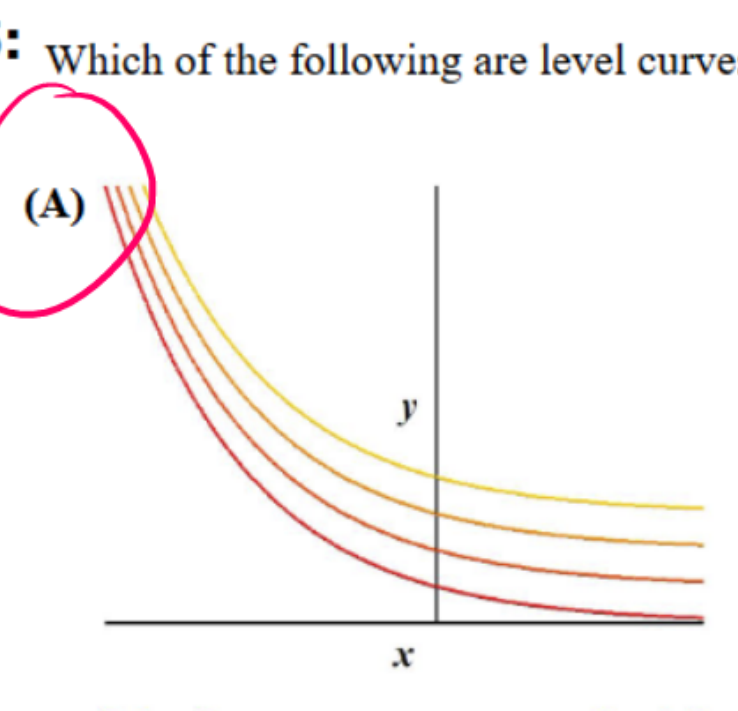
$= \frac{(7)(1) + 0}{(7)(0) - 2(1)} = \boxed{\frac{-7}{2}} \qquad = 7$

b) $x = r\cos\theta = 0 \quad y = r\sin\theta = 2$

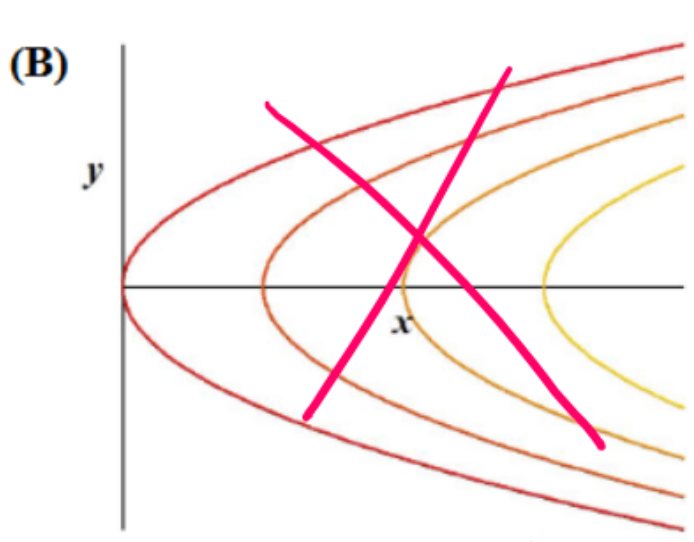
$2 = -7/2(0) + b \Rightarrow \boxed{b = 2}$

Problem #5: Which of the following are level curves for the function $f(x, y) = \ln(y - e^{-x})$?

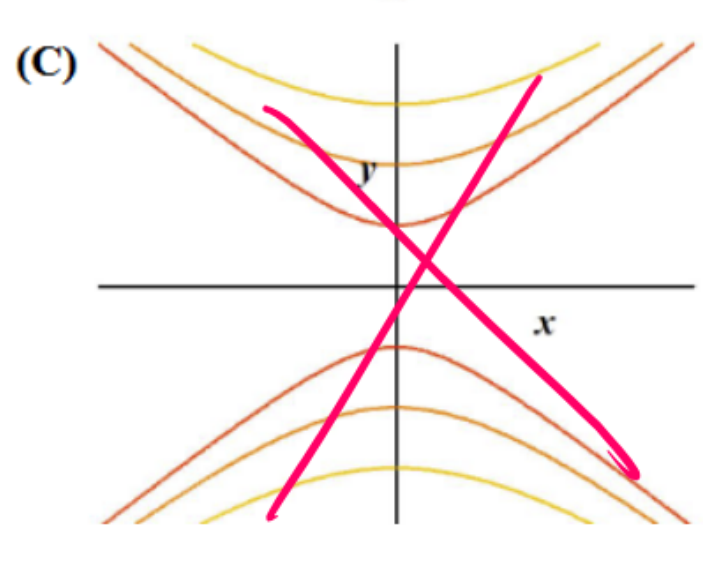
(A)



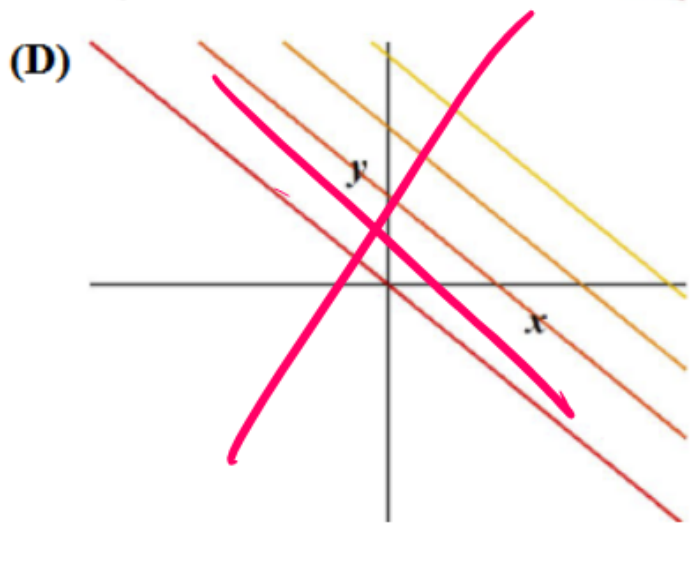
(B)



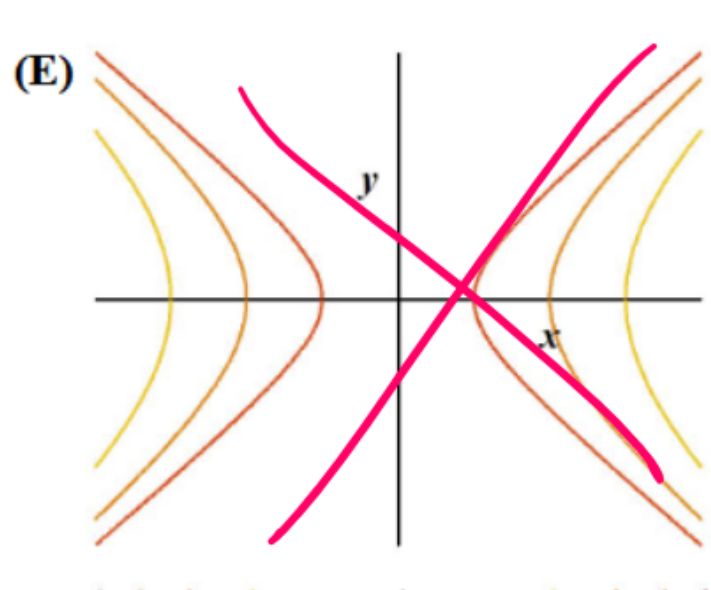
(C)



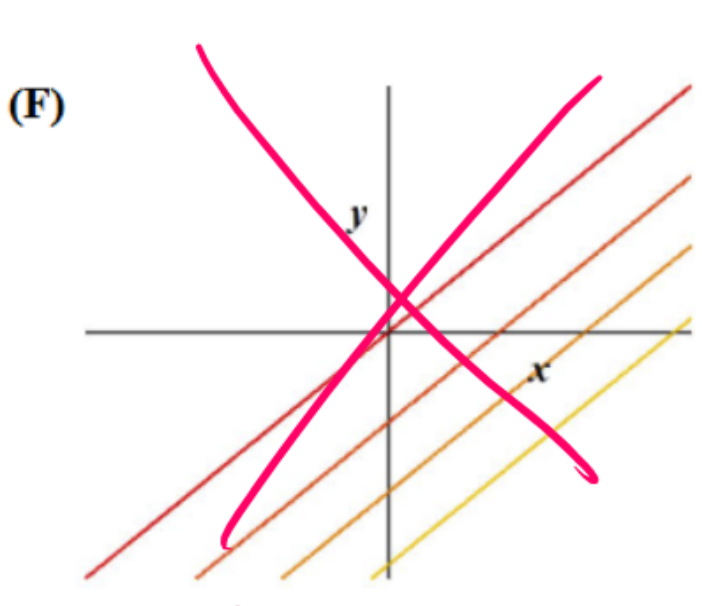
(D)



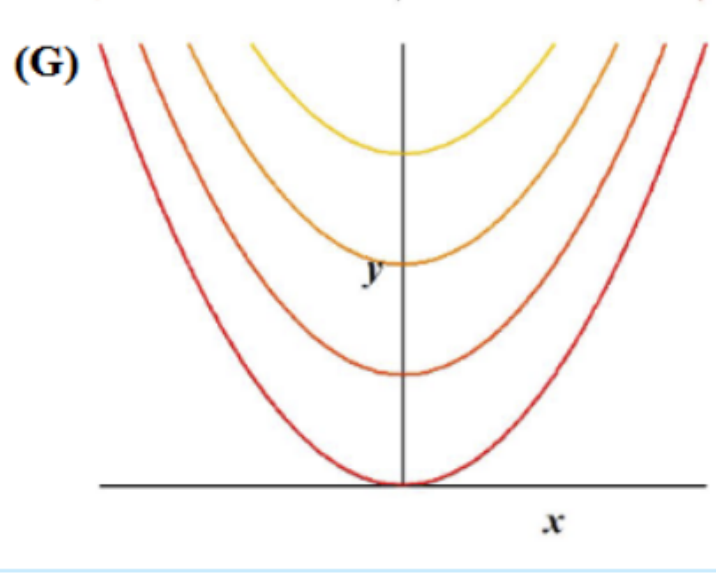
(E)



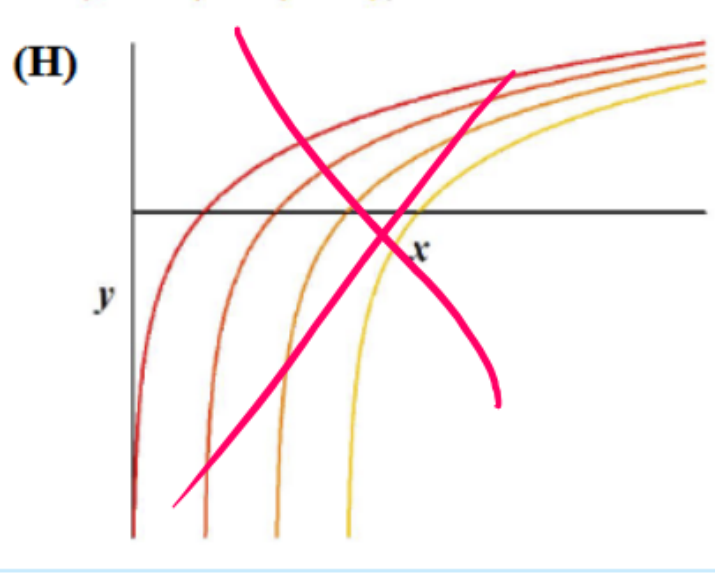
(F)



(G)



(H)



Consider $k = \ln(y - e^{-x})$

$\Rightarrow e^k = y - e^{-x}$

$\Rightarrow y = e^k + e^{-x}$

tells us: y is always positive.

when $x = 0$: $y = e^k + 1$

$\lim_{x\to\infty} e^k + e^{-x} = e^k + 1$ (constant)

$\lim_{x\to-\infty} e^k + e^{-x} = \infty$

Pick graph (A).

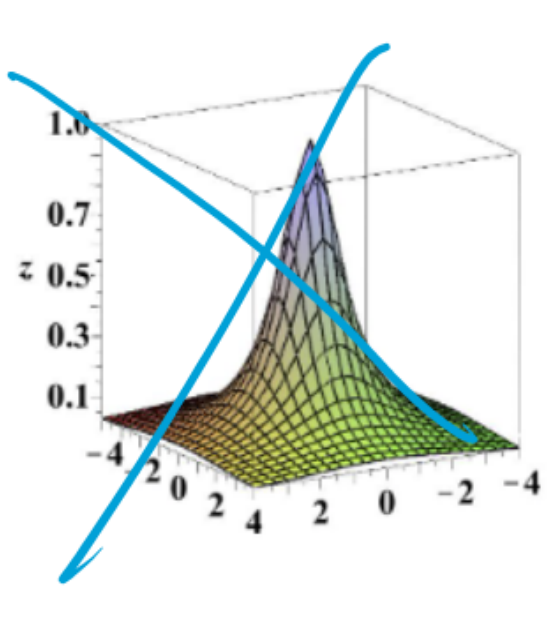
Problem #6: Sketch the graph of the function $f(x, y) = x - y^2$.

(treat $f(x,y) = z$) keeping x constant

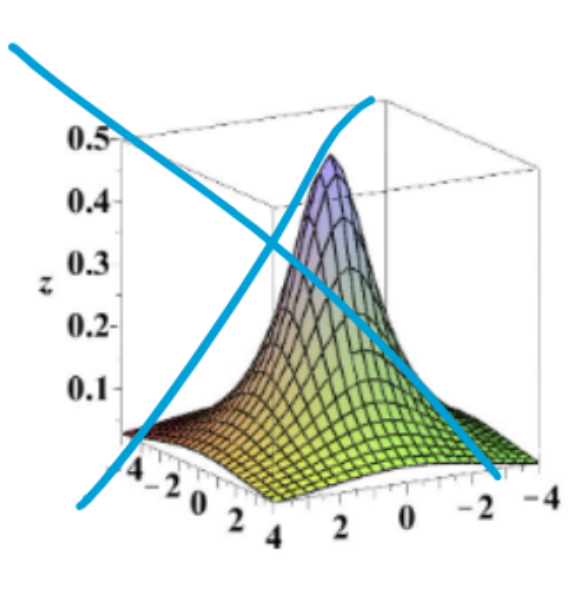
when $y = 0$ $f(x,y) = x$
when $y > 0$ $f(x,y)$ is decr.
when $y < 0$ $f(x,y)$ is still decr...

Seems like G is the closest.

(A)



(B)



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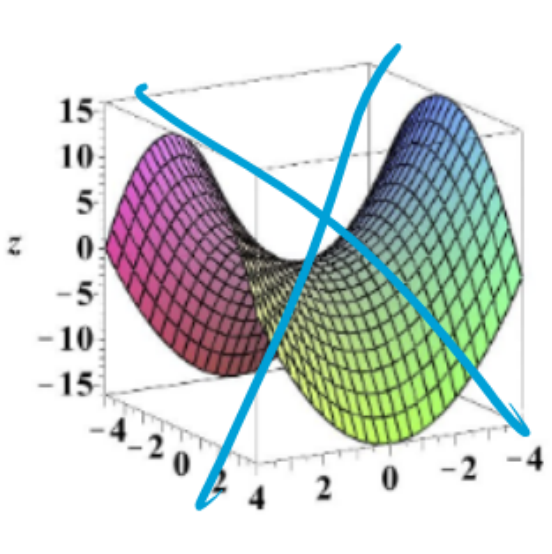

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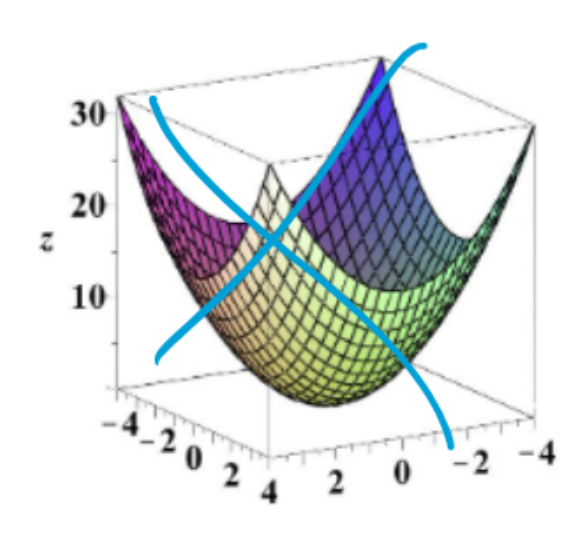

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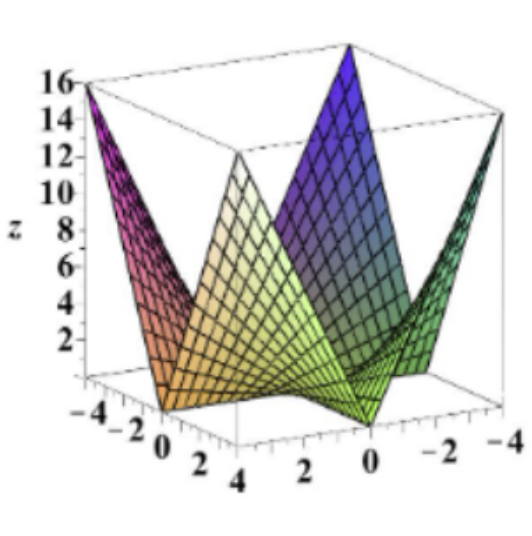

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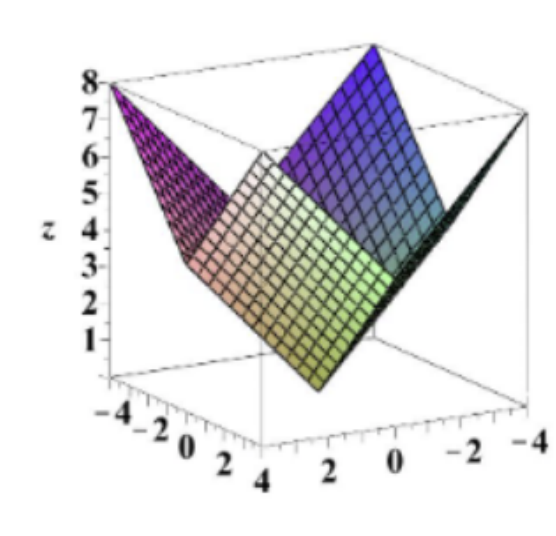

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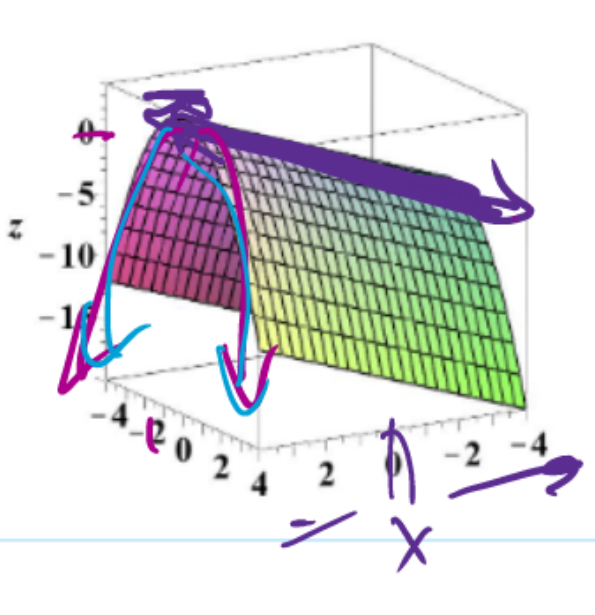

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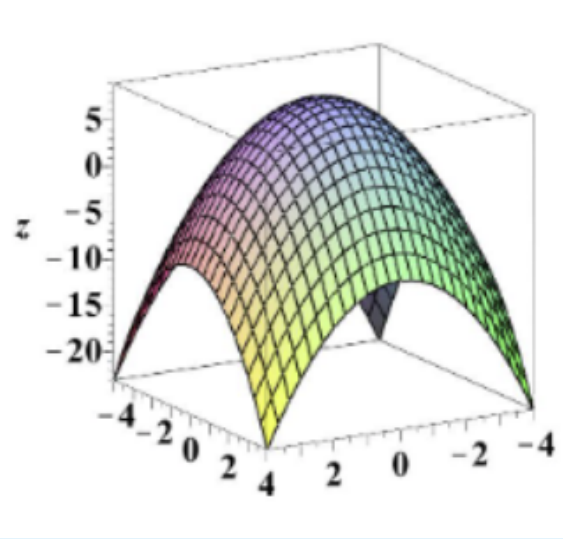


Problem #7: Which of the following functions is continuous at (0, 0)?

(i) $f(x,y) = \begin{cases} \frac{xy}{x^2+5y^2} & \text{if } (x,y) \ne (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$

(ii) $g(x,y) = \begin{cases} \frac{x^6y^6}{x^6+3y^6} & \text{if } (x,y) \ne (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$

(iii) $h(x,y) = \begin{cases} \frac{\sqrt{x^2+y^2+4}-2}{x^2+y^2} & \text{if } (x,y) \ne (0,0) \\ \frac{1}{2} & \text{if } (x,y) = (0,0) \end{cases}$

(i) if $y = 0$: $f(x,0) = \frac{0}{x^2} = 0$
if $x = 0$: $f(0,y) = \frac{0}{5y^2} = 0$
if $y = x$ $f(x,x) = \frac{x^2}{6x^2} = \frac{1}{6}$
doesn't match.

NOT CTS!

(ii) if $y = 0$: $g(x,0) = \frac{0}{x^6} = 0$. If $x = 0$: $g(0,y) = \frac{0}{3y^2} = 0$

if $y = kx$: $g(x, kx) = \frac{x^6k^6x^6}{x^6 + 3k^6x^6} = \frac{k^6x^{12}}{x^6(1+3k^6)} = \frac{k^6x^6}{(1+3k^6)} \xrightarrow{x\to0} 0$

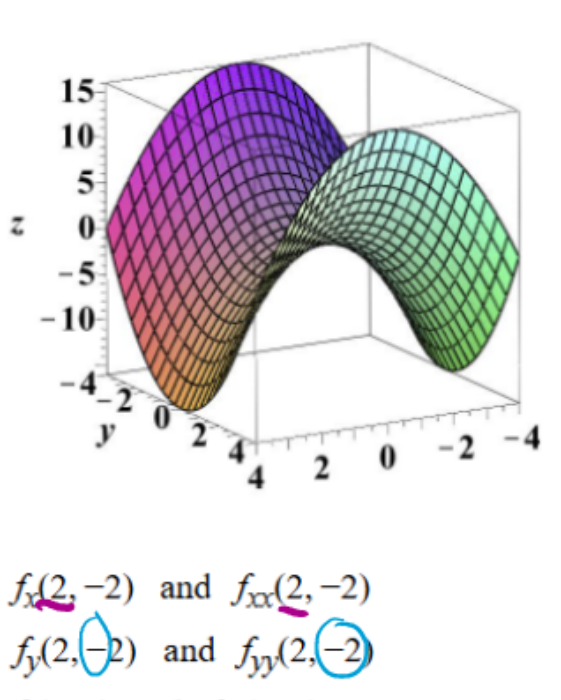
choice of k was arbitrary → CTS

(iii) if $x = y$: $h(x,x) = \frac{\sqrt{x^2+x^2+4}-2}{x^2+x^2} = \frac{\sqrt{2x^2+4}-2}{2x^2}\frac{(\sqrt{2x^2+4}+2)}{(\sqrt{2x^2+4}+2)}$

$= \frac{2x^2+4-4}{2x^2(\sqrt{2x^2+4}+2)} = \frac{\cancel{2x^2}}{\cancel{2x^2}(\sqrt{2x^2+4}+2)} = \frac{1}{\sqrt{2x^2+4}+2}$

when $x = 0$: $\frac{1}{\sqrt{4}+2} = \frac{1}{4} \ne \frac{1}{2}$ not continuous.

Problem #8: The graph of $z = f(x, y)$ is shown below. In each part, determine whether the given partial derivatives are positive, negative, or zero. (Note that the function is symmetric about 0 in both the x- and y- directions.)



(a) $f_x(2,-2)$ and $f_{xx}(2,-2)$
(b) $f_y(2,-2)$ and $f_{yy}(2,-2)$
(c) $f_x(0,2)$ and $f_{xx}(0,2)$
(d) $f_y(0,2)$ and $f_{yy}(0,2)$

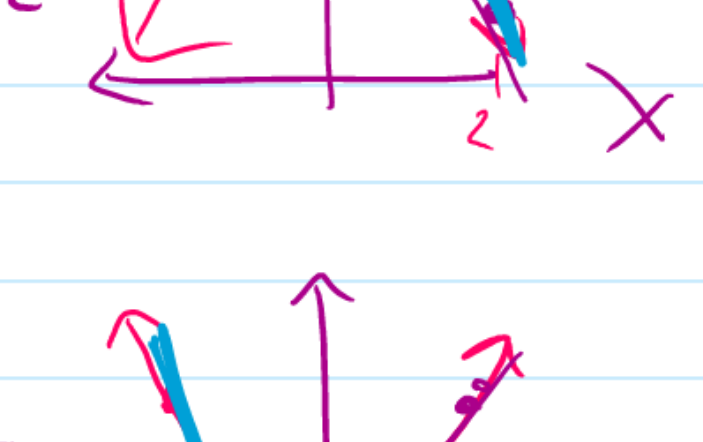
Recall:

$f_x := \frac{\partial f}{\partial x} \qquad f_{xx}: \frac{\partial^2 f}{\partial x^2}$

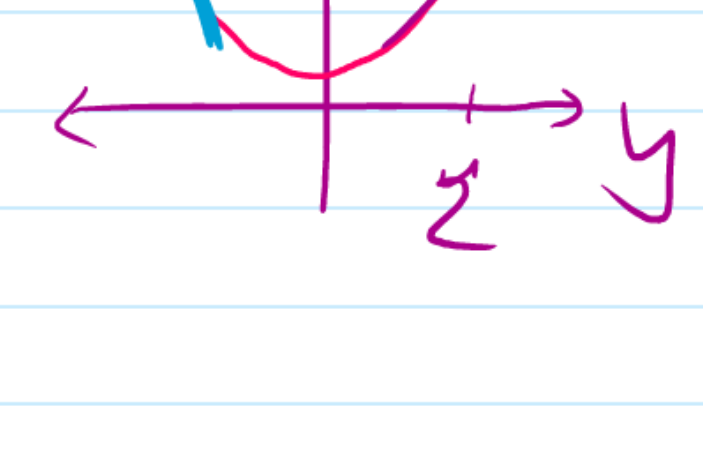
$f_y = \frac{\partial f}{\partial y} \qquad f_{yy}: \frac{\partial^2 f}{\partial y^2}$

incr/decr >0 <0 $\qquad$ concave up/down >0 <0

X side: z



Y side: z



a) negative, negative $\quad$ b) negative, positive

c) zero, negative $\quad$ d) positive, positive

(Warning: drop down options were So ANNOYING!)

Problem #9: Let $f(x,y,z) = y^{x/z}$. Find the value of the following partial derivatives.

(a) $f_x(3,4,3)$
(b) $f_y(4,4,2)$
(c) $f_z(3,2,4)$

$f = e^{\ln(y^{x/z})} = e^{\frac{x}{z}\ln y)}$

Part a)

$\frac{\partial f}{\partial x} = e^{\frac{x}{z}\ln y)} \cdot \frac{1}{z}\ln y) \xrightarrow{(3,4,3)} e^{\ln(4)} \cdot \frac{1}{3}\ln(4)$

Part b)

$\frac{\partial f}{\partial y} = \left(\frac{x}{z}\right)y^{\left(\frac{x}{z}-1\right)} \xrightarrow{(4,4,2)} \left(\frac{4}{2}\right)(4)^{\left(\frac{4}{2}-1\right)} = 2(4) = 8$

Part c)

$\frac{\partial f}{\partial z} = e^{\frac{x}{z}\ln y)}(-1)z^{-2}(x\ln y) \xrightarrow{(3,2,4)} e^{\frac{3}{4}\ln(2)}\frac{(-1)}{4^2}(3\ln(2))$

$= -\frac{3}{16}e^{3/4\ln(2)}\ln(2)$